

RESEARCH DESIGN

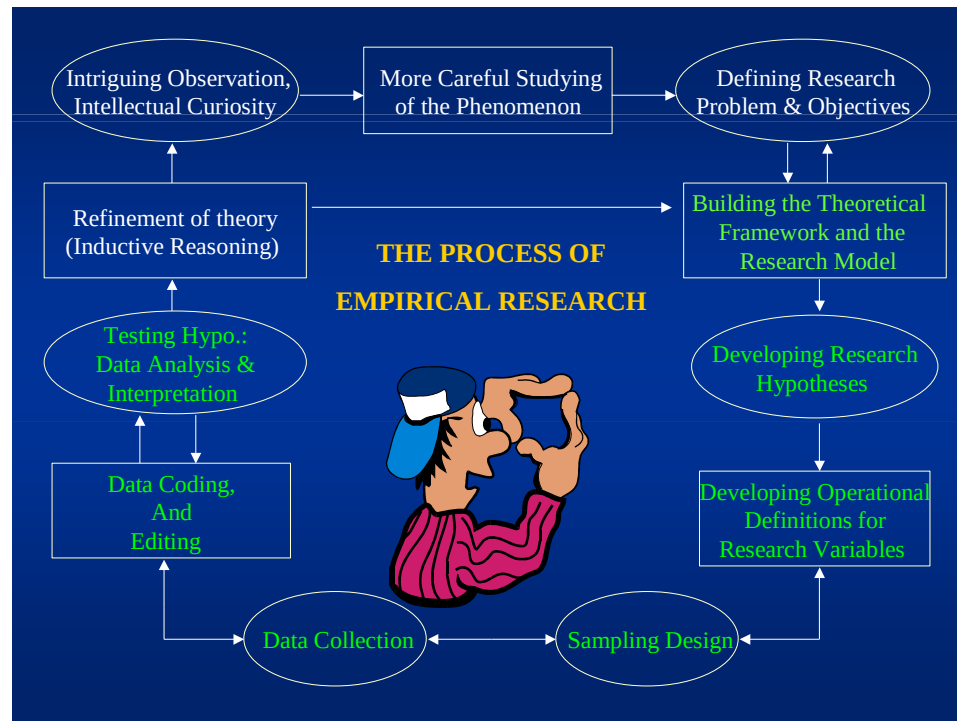
PROCESS OF DESIGNING AND CONDUCTING A RESEARCH PROJECT:

- What--What was studied?
- What about--What aspects of the subject were studied?
- What for--What is/was the significance of the study?
- What did prior lit./research say?
- What was done--How was the study conducted?
- What was found?
- So what?
- What now?

1. Introduction, Research Problems/ Objectives, & Justification
2. Literature Review
3. Methodology (Research sample, data collection, measurement, data analysis)
4. Results & Discussion
5. Implications
6. Conclusions and Recommendations for Future Research



➤ **RESEARCH DESIGN** refers to the plan, structure, and strategy of research--the blueprint that will guide the research process.





RESEARCH DESIGN: The blueprint/roadmap that will guide the research.

The test for the quality of a study's research design is the study's **conclusion validity**.

➤ **CONCLUSION VALIDITY** refers to the extent of researcher's ability to draw accurate conclusions from the research. That is, the degree of a study's:

a) **Internal Validity**—correctness of conclusions regarding the relationships among variables examined

- Whether the research findings accurately reflect how the research variables are really connected to each other.

b) **External Validity**—Generalizability of the findings to the intended/appropriate population/setting

- Whether appropriate subjects were selected for conducting the study



How do you achieve **internal and external validity** (i.e., conclusion validity)?

➤ By effectively controlling 3 types of **variances**:

- Variance of the INDEPENDENT & DEPENDENT variables (**Systematic Variance**)
- Variability of potential NUISANCE/EXTRANEIOUS/ CONFOUNDING variables (**Confounding Variance**)
- Variance attributable to ERROR IN MEASUREMENT (**Error Variance**).

How?

A cartoon character with an orange body, a red cap, and yellow shoes is running while holding a magnifying glass. Yellow starburst lines radiate from the character's head.

Effective Research Design

- **Guiding principle** for effective control of variances (and, thus, effective research design) is:

The **MAXMINCON** Principle

- **MAX**imize Systematic Variance
- **MIN**imize Error Variance
- **CON**trol Variance of Nuisance/Extraneous/Exogenous/Confounding variables

Effective Research Design

MAXimizing Systematic Variance:



Widening the range of values of research variables.

➤ IN EXPERIMENTS?

(where the researcher actually manipulates the independent variable and measures its impact on the dependent variable):

- Proper manipulation of experimental conditions to ensure high variability in indep. var.

➤ IN NON-EXPERIMENTAL STUDIES?

(where independent and dependent variables are measured simultaneously and the relationship between them are examined):

- Appropriate subject selection (selecting subjects that are sufficiently different with respect to the study's main var.)-- avoid **Range Restriction**

Effective Research Design

MINimizing Error Variance (measurement error):

Minimizing the part of variability in scores that is caused by error in measurement.

➤ **Sources of error variance:**

- Poorly designed measurement instruments (instrumentation error)
- Error emanating from study subjects (e.g., response error)
- Contextual factors that reduce a sound/accurate measurement instrument's capacity to measure accurately.

➤ **How to Minimize Error Variance?**

- Increase validity and reliability of measurement instruments.
- Measure variables under as ideal conditions as possible.





Effective Research Design

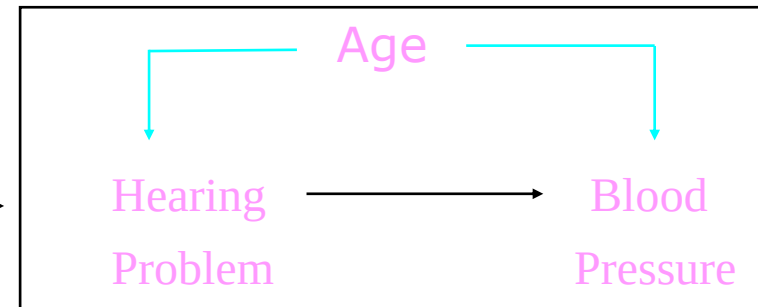
CONtrolling Variance of Confounding/Nuisance Variables:

FIRST, what are **Nuisance/Confounding** Variables?

- ✓ May or may not be of primary interest to the researcher,
- ✓ But, can produce undesirable variation in the study's dependent variable, and cause misleading or weird results
- ✓ Thus, if not controlled, can contaminate/distort the true relationship(s) between the independent and dependent variable(s) of interest
 - i.e., confounding var. can result in a **spurious**-- as opposed to **substantive**-- correlation between IV and DV. Example?

1. Historical data on **pollution** and **longevity**

2. Relationship between likelihood of **hearing problems** and **high blood pressure**



3. Recent stat. show **in-vitro** kids are 5 times more likely to develop **eye tumors** (Culprit: in-vitro fathers' older age)

4. Significantly more **armed store robberies** during the **cold winter days**.

Effective Research Design

HOW TO CONTROL FOR CONFOUNDING/ NUISANCE VARIABLES?



➤ **In Experimental Settings** (e.g., Fertilizer Amount → Rate of Plant Growth) :

Some Potential Confounding Variables?

- Conducting the experiment in a controlled environment (e.g., laboratory), where we can hold values of potential confounding variables constant.
- Subject selection (e.g., matching subjects in experiments)
- Random assignment of subjects (variations of confounding variables are evenly distributed between the experimental and control groups)

➤ **In Survey Research:**

- Sample selection (e.g., including only subjects with appropriate characteristics—using male college graduates as subjects will control for potential confounding effects of gender and education)
- **Statistical Control**--anticipating, measuring, and statistically controlling for confounding variables' effects (i.e., hold them statistically constant, or statistically removing their effects).

Effective Research Design

RECAP:

Effective research design is a function of ?

- ✓ Adequate (full range of) variability in values of research variables,
- ✓ Precise and accurate measurement,
- ✓ Identifying and controlling the effects of confounding variables,
and
- ✓ Appropriate subject selection



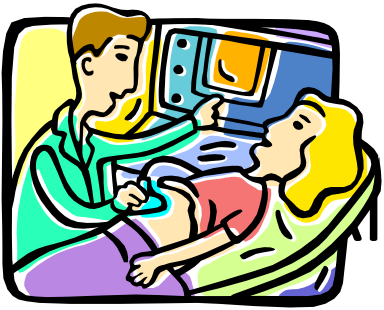


SPECIFIC TYPES OF RESEARCH DESIGN

BASIC RESEARCH DESIGNS:

- **Experimental Designs:**
 - True Experimental Studies
 - Pre-experimental Studies
 - Quasi-Experimental Studies
- **Non-Experimental Designs:**
 - Expost Facto/Correlational Studies

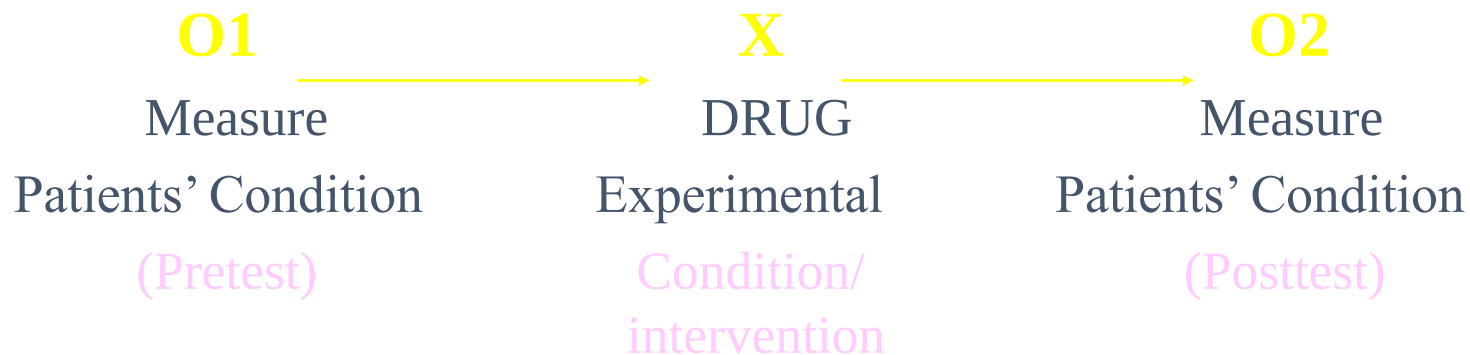




EXPERIMENTAL DESIGNS

One of the simplest experimental designs is the **ONE GROUP PRETEST-POSTTEST DESIGN--EXAMPLE?**

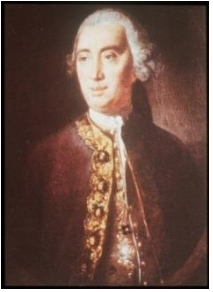
One way to examine :



- **RESULT:** Significant Improvement from O1 to O2 (i.e., sig. O2 - O1 difference)
- **QUESTION:** Did X (the drug) cause the improvement?

EXPERIMENTAL DESIGNS

would have been tempted to say “YES.”
He was a positivist and wanted to infer causality based on high correlations between events.



David Hume, 18th
Century Scottish
Philosopher

But such an inference could be seriously flawed.

Why?

- Have only shown “X” is a SUFFICIENT condition for the change “Y” (i.e., presence of X is associated with a change in Y).
 - But, is “X” also a NECESSARY condition for Y?
- How do you verify the latter?
 - By showing that the change would not have happened in the absence of X—using a CONTROL GROUP.



EXPERIMENTAL DESIGNS

- **CONTROL GROUP** simulates absence of X
 - Origin of using Control Groups (A tale from ancient Egypt)

Pretest Post-Test Control Group Design--Suppose random assignment (**R**) was used to control confounding variables:



- **RESULT:** $O2E > O1E$ & $O2C \text{ Not} > O1C$
QUESTION: Did X cause the improvement in Exp. Group?



EXPERIMENTAL DESIGNS

NOT NECESSARILY! Why not?

- Power of suggestibility (**The Hawthorne Effect**)

CONCLUSION?

- Need proper form of control—e.g., **Placebo**.



- **QUESTION:** Can we now conclude X caused the improvement in Exp. Group?
- **Maybe, but be aware of the Experimenter Effect** (it tends to prejudice the results especially in medical research).
- **SOLUTION: Double Blind Experiments** (neither the subjects nor the experimenter knows who is getting the placebo/drug).

EXPERIMENTAL DESIGNS

Experimental studies need to control for potential confounding factors that may threaten internal validity of the experiment:

- **Hawthorne Effect** is only one potential confounding factor in experimental studies.

Other such factors are:

- **History?**
 - Biasing events that occur between pretest and post-test
- **Maturation?**
 - Physical/biological/psychological changes in the subjects
- **Testing?**
 - Exposure to pretest influences scores on post-test
- **Instrumentation?**
 - Flaws in measurement instrument/procedure



EXPERIMENTAL DESIGNS

Experimental studies need to control for potential confounding factors that may threaten internal validity of the experiment (Continued):



- **Selection?**
 - Subjects in experimental & control groups different from the start
- **Statistical Regression (regression toward the mean)?**
 - Subjects selected based on extreme pretest values
 - Discovered by Francis Galton in 1877
- **Experimental Mortality?**
 - Differential drop-out of subjects from experimental and control groups during the study
- **Etc.**
- Experimental designs mostly used in natural and physical sciences.
- Generally, higher internal validity, lower external validity

TABLE 1
SOURCES OF INVALIDITY FOR DESIGNS 1 THROUGH 6

	Sources of Invalidity								
	Internal								External
	History	Maturation	Testing	Instrumentation	Regression	Selection	Mortality	Interaction of Selection and Maturation, etc.	Interaction of Testing and X
<u>Pre-Experimental Designs:</u>									
1. One-Shot Case Study X O	-	-				-	-		-
2. One-Group Pretest-Posttest Design O X O	-	-	-	-	?	+	+	-	-
3. Static-Group Comparison X O ----- O	+	?	+	+	+	-	-	-	-
<u>True Experimental Designs:</u>									
4. Pretest-Posttest Control Group Design R O X O R O O	+	+	+	+	+	+	+	+	-
5. Solomon Four-Group Design R O X O R O O R X O R O	+	+	+	+	+	+	+	+	+
6. Posttest-Only Control Group Design R X O R O	+	+	+	+	+	+	+	+	+

Note: In the tables, a minus indicates a definite weakness, a plus indicates that the factor is controlled, a question mark indicates a possible source of concern, and a blank indicates that the factor is not relevant.

It is with extreme reluctance that these summary tables are presented because they are apt to be "too helpful," and to be depended upon in place of the more complex and qualified presentation in the text. No + or - indicator should be respected unless the reader comprehends why it is placed there. In particular, it is against the spirit of this presentation to create uncomprehended fears of, or confidence in, specific designs.

SOURCES OF INVALIDITY FOR QUASI-EXPERIMENTAL DESIGNS 7 THROUGH 12

Sources of Invalidity												
Internal									External			
History	Maturation	Testing	Instrumentation	Regression	Selection	Mortality	Interaction of Selection and Maturation, etc.	Interaction of Testing and X	Interaction of Selection and X	Reactive Arrangements	Multiple-X Inference	
<u>Quasi-Experimental Designs:</u>												
7. Time Series O O O O X O O O O	-	+	+	?	+	+	+	+	-	?	?	
8. Equivalent Time Samples Design X ₁ O X ₂ O X ₃ O X ₄ O, etc.	+	+	+	+	+	+	+	+	-	?	-	-
9. Equivalent Materials Samples Design M ₁ X ₁ O M ₂ X ₂ O M ₃ X ₃ O M ₄ X ₄ O, etc.	+	+	+	+	+	+	+	+	-	?	?	-
10. Nonequivalent Control Group Design O X O O O	+	+	+	+	?	+	+	-	-	?	?	
11. Counterbalanced Designs X ₁ O X ₁ O X ₂ O X ₂ O X ₃ O X ₃ O X ₄ O X ₄ O X ₅ O X ₅ O X ₆ O X ₆ O X ₇ O X ₇ O X ₈ O X ₈ O	+	+	+	+	+	+	+	?	-	?	?	-
12. Separate-Sample Pretest-Posttest Design R O (X) R X O	-	-	+	?	+	+	-	-	+	+	+	
12a. R O (X) R X O R O (X) R X O	+	-	+	?	+	+	-	+	+	+	+	
12b. R O ₁ (X) R O ₂ (X) R X ₁ O ₁ X ₂ O ₂	-	+	+	?	+	+	-	?	+	+	+	
12c. R O ₁ X O ₂ R X O ₃	-	-	+	?	+	+	+	-	+	+	+	

CORRELATIONAL DESIGNS

NON-EXPERIMENTAL/CORRELATIONAL DESIGNS

- The design of choice in social sciences since the **phenomenon** under study is usually not reproducible in a laboratory setting
- Researcher has little or no control over study's indep., dep. and the **numerous potential confounding variables**,
- Often the researcher concomitantly measures all the study variables (e.g., independent, dependant, etc.),
- Then examines the following types of relationships:
 - correlations among variables or
 - differences among groups,
- Inability to control for effects of confounding variables makes causal inferences regarding relationships among variables more difficult and, thus:
- Generally, higher external validity, lower internal validity



CORRELATIONAL DESIGNS



Non-experimental designs rely on correlational evidence.

QUESTION: Does a significant correlation between two variables in a non-experimental study necessarily represent a causal relationship between those variables?

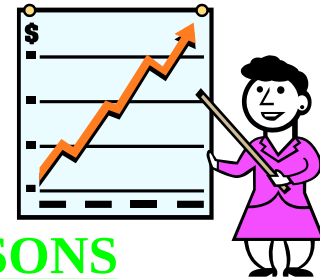
- **NOT NECESSARILY! EXAMPLES:**

- Water Fluoridation and AIDS
(*San Francisco Chronicle*, Sep. 6, 1984)
- Armed store robberies and cold weather
- Longevity and Pollution
- In-vitro birth and likelihood of developing eye tumors
- Hearing problem and blood pressure



- What can a significant correlation mean then?

CORRELATIONAL STUDIES



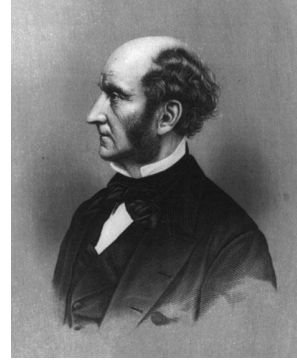
AT LEAST FOUR OTHER POSSIBLE INTERPRETATIONS/REASONS **FOR CORRELATIONS BETWEEN TWO VARIABLES:**

- a. Both variables are effects of a common cause (or both correlated with a third variable), i.e., spurious correlation
(e.g., air pollution and life expectancy, hearing problem & blood pressure, country's annual ice cream sales and frequency of hospital admissions for heat stroke)
- b. Both var. alternative indicators of same concept
(e.g., Church attend. & Freq. of Praying--religiosity).
- c. Both parts of a common "system" or "complex;" tend to come as a package
(e.g., martini drinking and opera attendance--life style)
- d. Fortuitous--Coincidental correlation, no logical relationship
(e.g., Outcome of super bowl games and movement of stock market)

CORRELATIONAL STUDIES

WHEN IS IT SAFER TO INFER CAUSAL LINKAGES FROM STRONG CORRELATIONS?

John Stuart Mill's Rules for Inferring Causal Links:



John Stuart Mill
1806-1873

- **Covariation Rule** (X and Y must be correlated)--Necessary but not sufficient condition.
- **Temporal Precedence Rule** (If X is the cause, Y should not occur until after X).
- **Internal Validity Rule** (Alternative plausible explanations of Y and X-Y relationships should be ruled out (i.e., eliminate other possible causes).
 - In practice, this means exercising caution by identifying potential confounding variables and controlling for their effects).